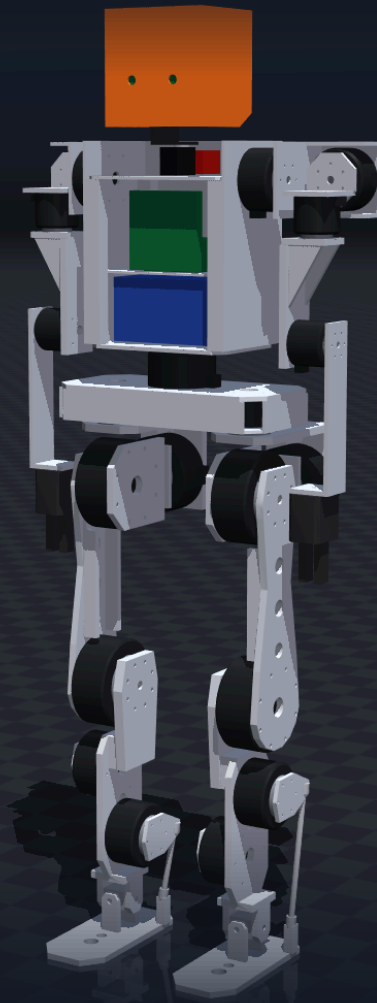


JX1

JNTUH's first humanoid robot



1.23 m

HEIGHT

33.6 kg

MASS

23

JOINTS (DOF)

21

ACTUATORS

50

BOM LINES

ESTIMATED COST OF THE ROBOT

₹7,12,481

WITH 15 % PROTOTYPE CONTINGENCY

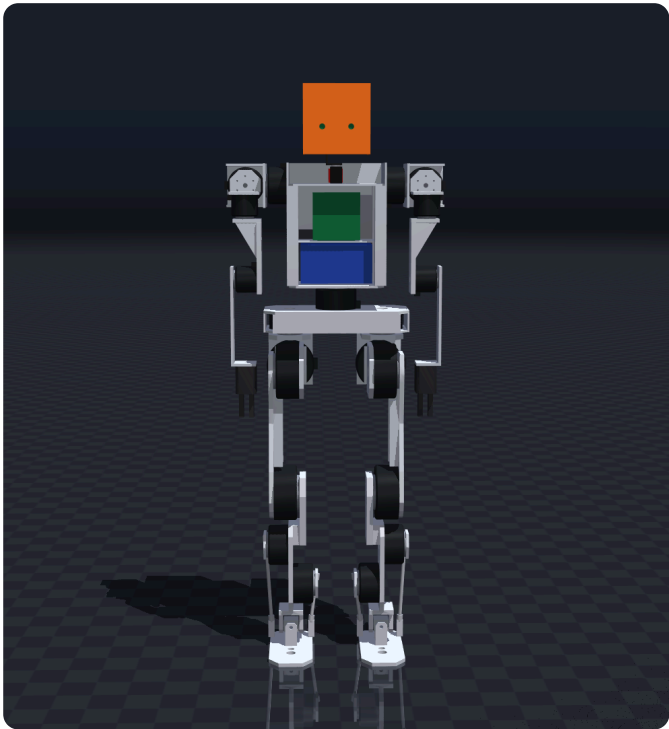
₹8,19,353

1. The JX1 humanoid

JX1 is a full-size walking humanoid robot designed to be built from parts that can be bought in India and metal plates that any laser-cutting or CNC job shop can make. The complete design (3D CAD, strength analysis, wiring, firmware, physics simulation and a trained walking controller) is finished and verified on the computer. This document lists every component of the first prototype and what it costs.

Specification

Height / mass	1.23 m / 33.6 kg (CAD, all parts)
Joints (DOF)	23 — legs 12 (6 per leg), waist 1, arms 8 (4 per arm), neck 2
Legs	thigh 0.27 m, shin 0.30 m, hip spacing 0.20 m, foot 210 × 95 mm; parallel ankle (2 motors per ankle)
Actuators	21 RobStride 48 V CAN actuators (RS04, RS03, RS06, RS02, RS00) + 2 neck bus servos
Brain	NVIDIA Jetson Nano 4 GB — runs the learned walking controller 50 times a second
Motor control	2 × Teensy 4.1 hub boards, 6 CAN buses at 1 Mbit/s, 500 Hz control loop
Sensors	BNO085 IMU, IMX219-83 stereo camera, 21 joint encoders, optional foot pressure sensors
Battery	13S2P Li-ion, 41.6–54.6 V, 468 Wh; ≈ 1.4 h walking at 0.5 m/s, ≈ 5 h standing (calculated)
Structure	6061-T6 / 7075-T6 aluminium plates, every structural part strength-checked (FEA, SF ≥ 1.5)
Walking	up to 0.8 m/s with the reinforcement-learning controller (simulation of the CAD model)



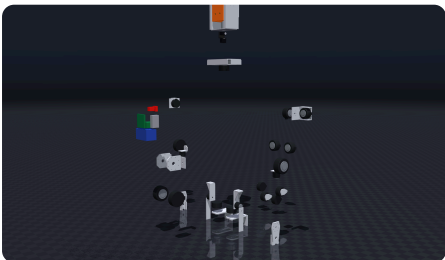
Design status

- ✓ 3D CAD of all 87 parts; every joint moved through its range and checked for collisions
 - ✓ Strength analysis (FEA) of every structural part — all pass in aluminium
 - ✓ Physics simulation: standing, squatting, walking, push recovery
 - ✓ Walking controller trained (reinforcement learning), tested in simulation and over ROS 2
 - ✓ Drawings and STEP files for the job shop, wiring diagrams, hub firmware
- Next: supplier quotes → purchase → assembly → first power-on

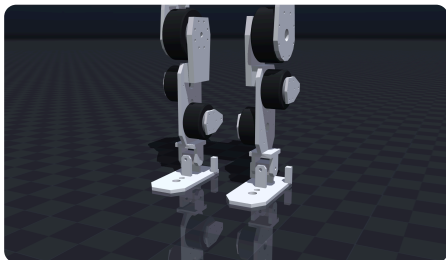
How this estimate was made

- Indian catalogue prices (Robu, Evelta, IndustryBuying, Moglix, bearinghouse.in ...), GST included where stated.
- Imported actuators: USD list price × ₹95.96 × 1.323 (import duty) × 1.07 (shipping) = landed cost.
- Machined and laser-cut parts: ESTIMATED from job-shop rates and plate prices — quotes required.
- 15 % contingency for a first prototype (rework, spares, price changes).

From CAD to a walking robot



87 parts, exploded view of the CAD



assembly step 11 of 22: ankles and feet



walking in the physics simulation

2. Cost summary

ROBOT TOTAL

₹7,12,481

49 line items

CONTINGENCY 15 %

₹1,06,872

first-prototype allowance

TOTAL BUDGET

₹8,19,353

≈ ₹8.19 lakh

By type of component



The 21 joint actuators are **77 %** of the cost. Everything else — metal, electronics, battery — together is ₹1,66,483.

By subsystem

SUBSYSTEM	COST	SHARE
Legs	₹4,18,394	58.7 %
Arms	₹1,52,688	21.4 %
Electronics	₹49,048	6.9 %
Power	₹31,606	4.4 %
Torso	₹31,522	4.4 %
Manufacturing	₹12,723	1.8 %
Head	₹11,406	1.6 %
Pelvis	₹5,094	0.7 %
Total	₹7,12,481	100 %

By build phase

PHASE	WHAT IT DELIVERS	COST
1	walking lower body: legs, pelvis, computer, electronics, battery, printing	₹5,13,833
2	upper body: waist, torso, arms, head, camera, foot sensors	₹1,98,648
3	grippers (deferred, not in the total)	₹7,000
Phases 1 + 2		₹7,12,481

Phase 1 alone produces a robot that stands and walks; it can be funded first.

Suggested purchase order

STAGE	WHAT IS BOUGHT	WHY	COST	CUMULATIVE
A	one actuator of each leg size (RS04, RS03, RS06), USB-CAN adapter, one Teensy 4.1, 2 kg PETG-CF	confirm bolt patterns, CAN control and fit-check prints before ordering metal	₹1,02,156	₹1,02,156
B	rest of phase 1: 9 more leg actuators, all metal leg parts, Jetson, hub PCB, battery, power	lower body stands and walks on a safety gantry	₹4,11,677	₹5,13,833
C	phase 2: waist, torso, 8 arm actuators, head, camera, foot sensors	complete humanoid	₹1,98,648	₹7,12,481
—	15 % contingency, released as needed	rework, spares, price changes	₹1,06,872	₹8,19,353

3. Joint actuators — the largest cost

Each actuator is a complete joint: brushless motor, planetary gearbox, absolute encoder and motor driver in one housing, controlled over CAN at 48 V. They are bought from RobStride (not stocked by Indian distributors) and imported; the prices below are landed costs.

ID	MODEL	USED FOR	TORQUE PEAK / RATED	KG	QTY	UNIT	TOTAL
001	RobStride RS04	hip pitch ×2, knee ×2	120/40 N·m	1.42	4	₹34,634	₹1,38,536
002	RobStride RS03	hip roll ×2	60/20 N·m	0.88	2	₹30,560	₹61,120
003	RobStride RS06	hip yaw ×2, ankle ×4 (2 per ankle)	36/11 N·m	0.62	6	₹28,522	₹1,71,132
004	RobStride RS06	waist yaw	36/11 N·m	0.62	1	₹28,522	₹28,522
005	RobStride RS02	shoulder pitch ×2, shoulder roll ×2	17/6 N·m	0.38	4	₹19,694	₹78,776
006	RobStride RS00	shoulder yaw ×2, elbow ×2	14/5 N·m	0.31	4	₹16,978	₹67,912
21 actuators					21		₹5,45,998

Joint map

JOINT	ACTUATOR	COUNT
hip yaw · hip roll · hip pitch	RS06 · RS03 · RS04	2 · 2 · 2
knee	RS04	2
ankle (parallel, 2 motors each)	RS06	4
waist yaw	RS06	1
shoulder pitch · shoulder roll	RS02 · RS02	2 · 2
shoulder yaw · elbow	RS00 · RS00	2 · 2
neck yaw · pitch (see §5)	Waveshare ST3215 servo	2

Option: China distributor

MODEL	QTY	LANDED	VIA CHINA	SAVING
RobStride RS04	4	₹34,634	₹24,008	₹42,504
RobStride RS03	2	₹30,560	₹20,004	₹21,112
RobStride RS06	6	₹28,522	₹17,000	₹69,132
RobStride RS06	1	₹28,522	₹17,000	₹11,522
RobStride RS02	4	₹19,694	₹13,997	₹22,788
RobStride RS00	4	₹16,978	₹11,974	₹20,016
All 21		₹5,45,998	₹3,58,924	₹1,87,074

Same units priced from the Chinese list (CNY, ₹13.52/CNY) plus a 5 % buying-agent fee, duty and shipping.

Are these actuators strong enough?

JOINT	ACTUATOR	PEAK TORQUE WALKING 0.8 M/S	WITH MARGIN	AVAILABLE	USED
Hip yaw	RS06	8.3 N·m	12.5 N·m (1.5×)	36 N·m	35 %
Hip roll	RS03	32.2 N·m	48.3 N·m (1.5×)	60 N·m	81 %
Hip pitch	RS04	22.5 N·m	33.8 N·m (1.5×)	120 N·m	28 %
Knee	RS04	59.6 N·m	89.4 N·m (1.5×)	120 N·m	74 %
Ankle pitch	2 × RS06 via linkage	37.7 N·m	37.7 N·m (1.0×)	46 N·m	82 %
Ankle roll	2 × RS06 via linkage	17.6 N·m	17.6 N·m (1.0×)	51 N·m	35 %

Simulated on the full CAD model with the trained walking controller (calculated, not yet measured). Hip and knee peaks are multiplied by the 1.5 safety margin used throughout the design; ankle values are compared with what the two-motor linkage can deliver; the ankle motors are the tightest, and the hub firmware caps every motor at 80 % of its peak torque. The next smaller RobStride size would not be enough at the knee (89 N·m needed, 60 N·m available) or at hip roll (48 N·m needed, 36 N·m available).

4. Structure and mechanical parts

The skeleton is flat 6061-T6 aluminium plate (7075-T6 for the most loaded part, the hip-yaw bracket), laser-cut or CNC-machined from the STEP files and A3 drawings in the repository. 3D-printed versions were analysed too and failed: the load path must be metal.

ID	PART	MATERIAL AND PROCESS	KG	FEA SF	QTY	UNIT	TOTAL
007	Hip yaw bracket (L/R)	7075-T6 CNC from 25 mm plate (top 10, back 12, keel 12 mm), ~0.8 kg	0.80	1.77 / 1.58	2	₹6,500	₹13,000
008	Hip roll bracket (L/R)	6061-T6 10 mm laser/waterjet plates + 20×20 corner bar, drilled/tapped, dowelled, ~0.5 kg	0.50	2.76 / 3.80	2	₹1,800	₹3,600
009	Thigh (L/R)	6061-T6 10 mm plate + run-out flanges (CNC, 2 setups), ~0.87 kg	0.87	2.31 / 2.78	2	₹4,500	₹9,000
010	Shin (L/R)	6061-T6: 10 mm web + 14 mm knee plate (laser-cut) + machined joggle block + 9 mm fork tines, ~0.95 kg	0.95	3.37 / 4.17	2	₹3,200	₹6,400
011	Ankle cross (spider)	EN8/EN24 steel, 32×33×26 turned + 2 cross-drilled Ø8 bores	0.20	steel	2	₹800	₹1,600
014	Ankle crank	6061 6 mm plate, CNC/laser, pilot recess	0.05	14.4 / 24.7	4	₹600	₹2,400
017	Foot (L/R)	6061-T6 8 mm sole (laser-cut) + bolted clevis tines + rod posts, ~0.5 kg	0.50	3.50 / 2.03	2	₹1,500	₹3,000
021	Pelvis torsion box	6061-T6 laser-cut plates (top 6, walls/bottom 4 mm) + tapped 8×8 corner bars, ~1.1 kg	1.10	2.90 / 3.44	1	₹3,000	₹3,000
041	Torso frame + battery bay	6061-T6 laser-cut plates (bottom 5, sides/deck/back 4, tray 3 mm) + 4× 2020 extrusion posts, ~2.4 kg	2.40	4.44 / 3.59	1	₹3,000	₹3,000
042	Arm structures (shoulder-pitch U-bracket, shoulder-roll L-bracket, upper arm, forearm) ×2	6061-T6 laser-cut 5-10 mm plates (upper arm 10/8 mm + gussets), bolted, ~2.0 kg both arms	2.00	2.05–5.99 / 1.66–4.84	1	₹6,000	₹6,000
Metal parts							₹51,000

FEA SF = safety factor, static / fatigue (strength ÷ worst stress; 1.5 required). Prices ESTIMATED from 3-axis CNC rates (US\$20–38/h) and plate at ₹380–400/kg — send the STEP + PDF set to 2–3 job shops for quotes.

Ankle linkage, bearings and fasteners

ID	ITEM	MODEL AND SPECIFICATION	QTY	UNIT	TOTAL
012	Ankle pins	Astro hard-chrome Ø8 rod 1 m (cut to 4 pins) — Ø8 h6 hard-chrome carbon steel	1	₹280	₹280
013	Needle bearings	N2K HK0810 — 8×12×10 drawn-cup needle bearing	8	₹35	₹283
015	Rod-end bearings	N2K POS5 / PHS5 (M5) — M5 rod end, female/male	8	₹142	₹1,133
016	Push-rods	Robu 1000 mm Ø8 chrome-plated smooth rod (SKU 501422) — cut 2 × 183 + 2 × 95 mm (ball-centre lengths per CAD), ends tapped M5 × 12 for POS5 male rod ends	1	₹330	₹330
018	Sole pad material	4 mm rubber anti-skid mat (cut) — 4 mm rubber	1	₹548	₹548
020	Leg fasteners & inserts	Unbrako SHCS M3/M4/M5 packs + Quartz M3 heat-set inserts — per robot allowance	1	₹3,000	₹3,000
Hardware					₹5,574

Structure and mechanical parts

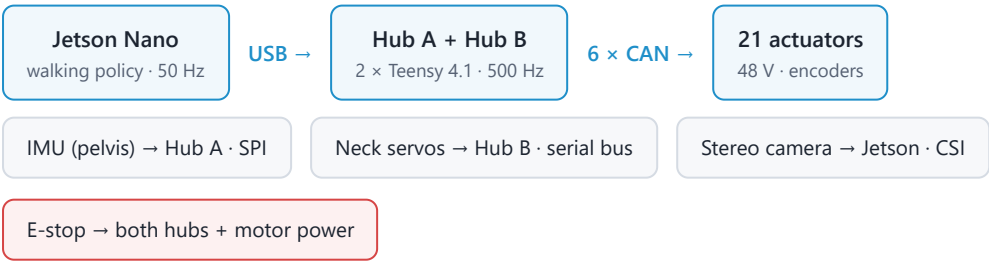
₹56,574

5. Electronics, computing and sensors

The Jetson Nano runs the walking controller; two Teensy 4.1 boards on a small carrier PCB talk to the actuators over six CAN buses, 500 times a second, and handle the emergency stop. The IMU senses balance; the stereo camera gives the robot depth vision.

ID	ITEM	SPECIFICATION	QTY	UNIT	TOTAL
019	Foot contact sensors SOUSHINE FSR406	39.7 mm FSR, 20 g-10 kg	8	₹379	₹3,032
022	IMU 7Semi BNO085 Nano breakout	9-DOF fusion IMU, SPI/I2C	1	₹2,094	₹2,094
023	High-level computer Waveshare JETSON-NANO-DEV-KIT (Nano 4 GB module, 16 GB eMMC, 64 GB card)	4× A57, 128-core Maxwell, 4 GB	1	₹30,900	₹30,900
024	Real-time CAN hub MCU PJRC Teensy 4.1	i.MX RT1062 600 MHz, 3× CAN (1× CAN-FD)	2	₹3,539	₹7,078
025	CAN transceivers NXP TJA1462AT (SOIC-8)	CAN/CAN-FD 8 Mbit/s SIC transceiver	8	₹121	₹966
026	CAN hub carrier PCB JX1 hub board (2× Teensy, 6× transceiver, E-stop input, 5 V/12 V rails)	2-layer 100×100 mm, 5 pcs	1	₹2,215	₹2,215
027	Bench USB-CAN adapter Waveshare USB-CAN-A	CAN 2.0A/B up to 1 Mbps	1	₹2,189	₹2,189
028	CAN / power harness JST-GH 1.25 2-4 pin leads, twisted pair, XT30 pairs	per robot	1	₹3,500	₹3,500
029	Wi-Fi card Intel AC8265 for Jetson Nano	802.11ac + BT	1	₹2,200	₹2,200
044	Neck servos Waveshare ST3215 serial bus servo	19.5 kg·cm @ 7.4 V, 12-bit encoder	2	₹2,159	₹4,318
045	Bus-servo driver SmartElex serial bus servo driver board		1	₹509	₹509
046	Stereo camera Waveshare IMX219-83 stereo	2× 8 MP, 83° FOV, CSI	1	₹6,399	₹6,399
Electronics, sensors, servos					₹65,400

How the electronics connect



Foot pressure sensors (line 019) are optional (phase 2). The hub PCB price is for 5 boards from Lion Circuits' public calculator; one board is used.

6. Battery and power system

A 13S2P pack of 28 Samsung 21700 cells (48 V nominal, 468 Wh) feeds the actuators through an anti-spark switch and 58 V-rated branch fuses; isolated and step-down supplies feed the computers and servos.

ID	ITEM	SPECIFICATION	QTY	UNIT	TOTAL
030	Li-ion cells Samsung INR21700-50S	5.0 Ah, 45 A pulse / 25 A cont., 69 g	28	₹363	₹10,176
031	BMS JBD SP14S004 smart BMS 10-14S 40 A	13S configured, Bluetooth, precharge-capable	1	₹2,199	₹2,199
032	Pack build + test service lionbattery.in pack/BMS test service	spot welding, capacity test	1	₹2,000	₹2,000
033	Battery connector Amass XT90-S anti-spark pair	90 A	1	₹212	₹212
034	Motor-bus main switch Flipsky Anti-Spark Switch Smart Enhanced 300A V2.0	12-84 V, 100 A cont., remote-able	1	₹7,499	₹7,499
035	Emergency stop Schneider XB2BS8442C	40 mm mushroom, twist release, 1NC	1	₹399	₹399
036	Branch fuses Littelfuse 0891030.NXS 30 A 58 V ATO	58 V rated (suits 54.6 V bus)	4	₹88	₹352
037	Fuse holders 30 A inline ATO holder, 12 AWG		4	₹190	₹760
038	Jetson 5 V supply Mean Well DDR-60L-5	18-75 V in, 5 V 10.8 A isolated	1	₹4,247	₹4,247
039	12 V auxiliary supply 6.5-60 V → adjustable 10 A synchronous buck	neck servos, fans	1	₹447	₹447
040	Charger 54.6 V 6 A Li-ion charger (13S)		1	₹3,315	₹3,315
Power system					₹31,606

Battery at a glance

Configuration	13S2P, 28 × 21700 cells
Voltage	41.6–54.6 V
Energy	468 Wh
BMS limit	40 A continuous (peak need 15.9 A)
Standing	73 W → 5.1 h
Walking 0.5 m/s	274 W → 1.37 h
Walking 0.8 m/s	347 W → 1.08 h

Safety built in

- Emergency stop with two independent paths: it cuts the 48 V motor bus *and* puts every joint into damping.
- 58 V-rated fuses on every branch (car fuses are rated 32 V and are not safe here).
- Smart BMS: over-charge, over-discharge and over-current protection, Bluetooth monitoring.
- XT90-S anti-spark service disconnect; pack built and capacity-tested by lionbattery.in.

7. 3D printing, consumables and optional items

ID	ITEM	USED FOR	QTY	UNIT	TOTAL
047	Head shell + neck brackets JX1 head	FDM PETG-CF ~0.3 kg (~12 h)	1	₹180	₹180
048	PA-CF filament eSUN PA-CF 1.75 mm 1 kg	neck bracket, grippers, small brackets (all structural parts are aluminium after FEA)	1	₹4,799	₹4,799
049	PETG-CF filament Numakers PETG-CF 1 kg	covers, head, battery box; PETG fit-check copies of the metal parts before ordering (~1.5 kg, ESTIMATED)	4	₹1,356	₹5,424
050	Misc. consumables cable ties, spiral wrap, heat-shrink, drag chain 10×10, thread-locker		1	₹2,500	₹2,500

Printing and consumables	₹12,903
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Print PETG **fit-check copies** of every metal part first and bolt them to the real actuators before ordering metal. Deferred to phase 3 (not in the total): simple grippers ×2 (₹7,000).

8. Ways to reduce the cost

OPTION	TRADE-OFF	CHANGE
Buy the same 21 RobStride actuators through a China distributor (CNY list + 5 % agent)	same parts; longer lead time, agent risk	-₹1,87,074
Motor-bus switch as a MOSFET stage on the hub PCB instead of the Flipsky switch	needs design and testing	-₹5,999
IMX219-77 mono camera instead of the IMX219-83 stereo camera	no stereo depth (walking unaffected)	-₹4,699
Synchronous buck instead of the isolated Mean Well 5 V supply	not isolated; more noise on the Jetson rail	-₹3,800
6061-T6 box-beam hip-yaw bracket (variant Y7) instead of CNC 7075-T6	hip stack +18 mm; re-verify CAD/FEA	-₹2,000
8 mm thigh plate (variant V7, SF 1.64 / 1.94)	smaller strength margin	-₹1,000
Printed TPU 95A sole instead of the rubber mat	grip to be tested	-₹148
All options together	budget build ≈ ₹5,07,761 (+ contingency ₹5,83,925)	-₹2,04,720

Upgrade path (not included): Jetson Orin Nano Super dev kit (₹50,000, +₹19,100) for more computing power and native CAN.

Not included in this estimate — budget separately

- Hand tools: hex keys, torque screwdriver and wrench, taps, reamers, calipers
 - Lifting gantry or chest harness for safe first power-on and walking tests
 - Job-shop setup charges above the estimate, customs-clearance fees, courier delays
- Bench equipment: current-limited power supply, multimeter, soldering station
 - Spares (e.g. one actuator of each size) and a second battery pack
 - A laptop/PC with a GPU if new walking controllers are to be trained

9. Complete list of components and total cost

#	GROUP	ITEM — MODEL	QTY	UNIT ₹	TOTAL ₹
001	Legs	Joint actuator — RobStride RS04	4	34,634	1,38,536
002	Legs	Joint actuator — RobStride RS03	2	30,560	61,120
003	Legs	Joint actuator — RobStride RS06	6	28,522	1,71,132
004	Torso	Joint actuator — RobStride RS06	1	28,522	28,522
005	Arms	Joint actuator — RobStride RS02	4	19,694	78,776
006	Arms	Joint actuator — RobStride RS00	4	16,978	67,912
007	Legs	Hip yaw bracket (L/R) — JX1_HipYawBracket	2	6,500	13,000
008	Legs	Hip roll bracket (L/R) — JX1_HipRollBracket	2	1,800	3,600
009	Legs	Thigh (L/R) — JX1_Thigh	2	4,500	9,000
010	Legs	Shin (L/R) — JX1_Shin	2	3,200	6,400
011	Legs	Ankle cross (spider) — JX1_AnkleCross	2	800	1,600
012	Legs	Ankle pins — Astro hard-chrome Ø8 rod 1 m (cut to 4 pins)	1	280	280
013	Legs	Needle bearings — N2K HK0810	8	35	283
014	Legs	Ankle crank — JX1_AnkleCrank	4	600	2,400
015	Legs	Rod-end bearings — N2K POS5 / PHS5 (M5)	8	142	1,133
016	Legs	Push-rods — Robu 1000 mm Ø8 chrome-plated smooth rod (S...	1	330	330
017	Legs	Foot (L/R) — JX1_Foot	2	1,500	3,000
018	Legs	Sole pad material — 4 mm rubber anti-skid mat (cut)	1	548	548
019	Legs	Foot contact sensors — SOUSHINE FSR406	8	379	3,032
020	Legs	Leg fasteners & inserts — Unbrako SHCS M3/M4/M5 packs + Quartz M3 hea...	1	3,000	3,000
021	Pelvis	Pelvis torsion box — JX1_Pelvis	1	3,000	3,000
022	Pelvis	IMU — 7Semi BNO085 Nano breakout	1	2,094	2,094
023	Electronics	High-level computer — Waveshare JETSON-NANO-DEV-KIT (Nano 4 GB mo...	1	30,900	30,900
024	Electronics	Real-time CAN hub MCU — PJRC Teensy 4.1	2	3,539	7,078
025	Electronics	CAN transceivers — NXP TJA1462AT (SOIC-8)	8	121	966
026	Electronics	CAN hub carrier PCB — JX1 hub board (2× Teensy, 6× transceiver, E...	1	2,215	2,215
027	Electronics	Bench USB-CAN adapter — Waveshare USB-CAN-A	1	2,189	2,189
028	Electronics	CAN / power harness — JST-GH 1.25 2-4 pin leads, twisted pair, XT...	1	3,500	3,500
029	Electronics	Wi-Fi card — Intel AC8265 for Jetson Nano	1	2,200	2,200
030	Power	Li-ion cells — Samsung INR21700-50S	28	363	10,176
031	Power	BMS — JBD SP14S004 smart BMS 10-14S 40 A	1	2,199	2,199
032	Power	Pack build + test service — lionbattery.in pack/BMS test service	1	2,000	2,000
033	Power	Battery connector — Amass XT90-S anti-spark pair	1	212	212
034	Power	Motor-bus main switch — Flipsky Anti-Spark Switch Smart Enhanced 30...	1	7,499	7,499
035	Power	Emergency stop — Schneider XB2BS8442C	1	399	399
036	Power	Branch fuses — Littelfuse 0891030.NXS 30 A 58 V ATO	4	88	352
037	Power	Fuse holders — 30 A inline ATO holder, 12 AWG	4	190	760
038	Power	Jetson 5 V supply — Mean Well DDR-60L-5	1	4,247	4,247
039	Power	12 V auxiliary supply — 6.5-60 V → adjustable 10 A synchronous buck	1	447	447
040	Power	Charger — 54.6 V 6 A Li-ion charger (13S)	1	3,315	3,315
041	Torso	Torso frame + battery bay — JX1_Torso	1	3,000	3,000
042	Arms	Arm structures (shoulder-pitch U-bracket, sho... — JX1_ShoulderPitchBracket / ShoulderRollBrac...	1	6,000	6,000
043	Arms	Simple grippers ×2 — JX1 parallel-jaw gripper (RS00-sized servo...	2	3,500	(7,000)
044	Head	Neck servos — Waveshare ST3215 serial bus servo	2	2,159	4,318
045	Head	Bus-servo driver — SmartElex serial bus servo driver board	1	509	509
046	Head	Stereo camera — Waveshare IMX219-83 stereo	1	6,399	6,399
047	Head	Head shell + neck brackets — JX1 head	1	180	180
048	Manufacturing	PA-CF filament — eSUN PA-CF 1.75 mm 1 kg	1	4,799	4,799
049	Manufacturing	PETG-CF filament — Numakers PETG-CF 1 kg	4	1,356	5,424
050	Manufacturing	Misc. consumables — cable ties, spiral wrap, heat-shrink, drag...	1	2,500	2,500

Robot total (49 lines; deferred item in brackets not included)	₹7,12,481
Prototype contingency 15 %	₹1,06,872
TOTAL ESTIMATED COST OF JX1	₹8,19,353
<i>Rupees Eight Lakh Nineteen Thousand Three Hundred Fifty-Three only</i>	